

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

In a scale model of an office building, a length of **1** inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. If a length in the actual office building measures x feet, which expression represents the corresponding length, in inches, in the model in terms of x ?

Answer

- A. $\frac{3x}{(17)(12)}$
- B. $\frac{17x}{(12)(3)}$
- C. $\frac{3x}{17}$
- D. $\frac{17x}{3}$

Correct Answer: C

Rationale

Choice C is correct. It’s given that in a scale model, a length of **1** inch is equivalent to a length of $\frac{17}{3}$ feet in the actual office building. This means that the ratio of a length, in feet, in the actual office building to the corresponding length, in inches, in the model is $\frac{17}{3}$ to **1**. Therefore, the length, in inches, in the model corresponding to a length in the actual office building of x feet can be found by dividing x by $\frac{17}{3}$. Dividing by $\frac{17}{3}$ is equivalent to multiplying by $\frac{3}{17}$, so the corresponding length, in inches, in the model in terms of x can be represented by the expression $\frac{3x}{17}$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This expression represents the corresponding length, in inches, in the model in terms of x for a model where **1** inch in the model is equivalent to a length of $\frac{3}{17}$, not $\frac{17}{3}$, feet in the actual building.